EXPERIMENT 9

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Design input forms that validate data (e.g., email, phone number) and display error messages using HTML/CSS, JavaScript (with Validator.js)

AIM:

The aim is to design input forms that validate data, such as email and phone number, and display error messages using HTML/CSS and JavaScript with Validator.js.

PROCEDURE:

Step 1: Setting Up the HTML Form Start by creating an HTML form with input fields for the email and phone number.

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Form Validation</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link href="https://fonts.googleapis.com/css2?family=Nunito:wght@400;600;700;800&display=swap" rel="stylesheet">
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="page-wrapper">
    <div class="card">
      <div class="banner">
        <h1>&#10022; Get in Touch</h1>
        <p>Fill in the form below &mdash; we'd love to hear from you!</p>
      </div>
 <div class="progress-row">
        <div class="pbar"><div class="pfill" id="pfill"></div></div>
        <span class="pct" id="pct">0%</span>
      </div>
      <form id="myForm" novalidate>
```

```
<div class="field-group">
  <label for="name">
    <span class="lico purple">1</span> Full Name
  </label>
  <div class="input-wrap">
    <svg class="field-icon" viewBox="0 0 20 20" fill="none">
      <circle cx="10" cy="7" r="3.5" stroke="#7c3aed" stroke-width="1.8"/>
      <path d="M3 17c0-3.314 3.134-6 7-6s7 2.686 7 6" stroke="#7c3aed" stroke-width="1.8" stroke-linecap="round"/>
    </svg>
    <input type="text" id="name" name="name" placeholder="e.g. Priya Sharma" autocomplete="off">
    <svg class="tick-icon" id="tick-name" viewBox="0 0 20 20" fill="none">
      <circle cx="10" cy="10" r="9" fill="#10b981"/>
      <path d="M6 10l3 3 5-5" stroke="#fff" stroke-width="2" stroke-linecap="round" stroke-linejoin="round"/>
    </svg>
  </div>
  <span class="msg" id="nameMsg"></span>
</div>
<div class="field-group">
  <label for="email">
    <span class="lico pink">2</span> Email Address
  </label>
  <div class="input-wrap">
    <svg class="field-icon" viewBox="0 0 20 20" fill="none">
      <rect x="2" y="5" width="16" height="12" rx="2" stroke="#db2777" stroke-width="1.8"/>
      <path d="M2 7l8 5 8-5" stroke="#db2777" stroke-width="1.8" stroke-linecap="round"/>
    </svg>
    <input type="text" id="email" name="email" placeholder="priya@example.com" autocomplete="off">
    <svg class="tick-icon" id="tick-email" viewBox="0 0 20 20" fill="none">
      <circle cx="10" cy="10" r="9" fill="#10b981"/>
      <path d="M6 10l3 3 5-5" stroke="#fff" stroke-width="2" stroke-linecap="round" stroke-linejoin="round"/>
    </svg>
  </div>
  <span class="msg" id="emailMsg"></span>
</div>
<div class="field-group">
  <label for="phone">
```

```
    <span class="lico amber">3</span> Phone Number
  </label>
  <div class="input-wrap">
    <svg class="field-icon" viewBox="0 0 20 20" fill="none">
      <rect x="5" y="2" width="10" height="16" rx="2" stroke="#f59e0b" stroke-width="1.8"/>
      <circle cx="10" cy="15" r="1" fill="#f59e0b"/>
    </svg>
    <input type="text" id="phone" name="phone" placeholder="+91 98765 43210" autocomplete="off">
    <svg class="tick-icon" id="tick-phone" viewBox="0 0 20 20" fill="none">
      <circle cx="10" cy="10" r="9" fill="#10b981"/>
      <path d="M6 10l3 3 5-5" stroke="#fff" stroke-width="2" stroke-linecap="round" stroke-linejoin="round"/>
    </svg>
  </div>
  <p class="hint">Include country code (e.g. +91 for India, +1 for US)</p>
  <span class="msg" id="phoneMsg"></span>
</div>
<div class="field-group">
  <label for="url">
    <span class="lico green">4</span> Website <span class="optional">optional</span>
  </label>

  <div class="input-wrap">
    <svg class="field-icon" viewBox="0 0 20 20" fill="none">
      <circle cx="10" cy="10" r="8" stroke="#10b981" stroke-width="1.8"/>
      <path d="M10 2c-2 3-2 13 0 16M10 2c2 3 2 13 0 16M2 10h16" stroke="#10b981" stroke-width="1.5" stroke-
linecap="round"/>
    </svg>
    <input type="text" id="url" name="url" placeholder="https://yoursite.com" autocomplete="off">
    <svg class="tick-icon" id="tick-url" viewBox="0 0 20 20" fill="none">
      <circle cx="10" cy="10" r="9" fill="#10b981"/>
      <path d="M6 10l3 3 5-5" stroke="#fff" stroke-width="2" stroke-linecap="round" stroke-linejoin="round"/>
    </svg>
  </div>
  <span class="msg" id="urlMsg"></span>
</div>
<button type="submit">Send Message &#10022;</button>
<div class="success-msg" id="successMsg">
```

```
        &#127881; Woohoo! Form submitted successfully &mdash; all fields are valid!
      </div>
    </form>
  </div>
 </div>
 <script src="https://cdnjs.cloudflare.com/ajax/libs/validator/13.11.0/validator.min.js"></script>
 <script src="script.js"></script>
</body>
</html>
```

Step 2:

Styling the Form with CSS Next, add some basic styling to make the form look nice.

CSS

```
* {
  box-sizing: border-box;
  margin: 0;
 padding: 0;
}
body {
  font-family: 'Nunito', Arial, sans-serif;
  min-height: 100vh;
  display: flex;
  justify-content: center;
  align-items: center;
  padding: 2rem 1rem;
  background: linear-gradient(135deg, #fdf4ff 0%, #eff6ff 50%, #f0fdf4 100%);
}
.page-wrapper {
  width: 100%;
  max-width: 480px;
}
.card {
  background: #ffffff;
  border-radius: 24px;
  padding: 2.2rem 2rem;
  border: 2px solid #f0e6ff;
```

```
  box-shadow: 0 8px 40px rgba(120, 60, 220, 0.10);
}
.banner {
  background: linear-gradient(135deg, #7c3aed 0%, #db2777 50%, #f59e0b 100%);
  border-radius: 16px;
  padding: 1.4rem 1.5rem;
  margin-bottom: 1.6rem;
  text-align: center;
}
.banner h1 {
  color: #ffffff;
  font-size: 24px;
  font-weight: 800;
  letter-spacing: -0.3px;
  margin-bottom: 4px;
}
.banner p {
  color: rgba(255, 255, 255, 0.88);
  font-size: 13px;
  font-weight: 600;
}
.progress-row {
  display: flex;
  align-items: center;
  gap: 10px;
  margin-bottom: 1.5rem;
}
.pbar {
  flex: 1;
  height: 7px;
  background: #ede9fe;
  border-radius: 99px;
  overflow: hidden;
}
.pfill {
```

```
  height: 100%;
  width: 0%;
  background: linear-gradient(90deg, #7c3aed, #db2777, #f59e0b);
  border-radius: 99px;
  transition: width 0.4s ease;
}

.pct {
  font-size: 12px;
  font-weight: 800;
  color: #7c3aed;
  min-width: 32px;
  text-align: right;
}
.field-group {
  margin-bottom: 1.2rem;
  display: flex;
  flex-direction: column;
}
label {
  display: flex;
  align-items: center;
  gap: 7px;
  font-size: 12px;
  font-weight: 800;
  letter-spacing: 0.6px;
  text-transform: uppercase;
  color: #6d28d9;
  margin-bottom: 7px;
}
.lico {
  width: 20px;
  height: 20px;
  border-radius: 50%;
  display: inline-flex;
```

```
  align-items: center;
  justify-content: center;
  font-size: 10px;
  font-weight: 800;
  color: #ffffff;
  flex-shrink: 0;
}
.lico.purple { background: #7c3aed; }
.lico.pink   { background: #db2777; }
.lico.amber  { background: #f59e0b; }
.lico.green  { background: #10b981; }
.optional {
  font-size: 11px;
  font-weight: 600;
  color: #c4b5fd;
  text-transform: none;
  letter-spacing: 0;
  margin-left: auto;
}
.input-wrap {
  position: relative;
}
.field-icon {
  position: absolute;
  left: 14px;
  top: 50%;
  transform: translateY(-50%);
  width: 18px;
  height: 18px;
  pointer-events: none;
}
.tick-icon {
  position: absolute;
  right: 13px;
  top: 50%;
```

```
  transform: translateY(-50%);
  width: 18px;
  height: 18px;
  display: none;
  pointer-events: none;
}
input {
  width: 100%;
  padding: 11px 44px 11px 44px;
  border: 2px solid #e9d5ff;
  border-radius: 12px;
  font-size: 15px;
  font-family: 'Nunito', Arial, sans-serif;
  font-weight: 600;
  color: #1e1b4b;
  background: #faf5ff;
  outline: none;
  transition: border-color 0.2s, box-shadow 0.2s, background 0.2s;
}
input::placeholder {
  color: #c4b5fd;
  font-weight: 600;
}
input:focus {
  border-color: #7c3aed;
  background: #ffffff;
  box-shadow: 0 0 0 4px rgba(124, 58, 237, 0.12);
}
input.valid {
  border-color: #10b981;
  background: #f0fdf4;
  box-shadow: 0 0 0 4px rgba(16, 185, 129, 0.10);
}
input.valid + .tick-icon {
  display: block;
```

```
}
input.invalid {
  border-color: #ef4444;
  background: #fff1f2;
  box-shadow: 0 0 0 4px rgba(239, 68, 68, 0.10);
}
.hint {
  font-size: 11px;
  color: #a78bfa;
  margin-top: 5px;
  font-weight: 600;
}
.msg {
  font-size: 12px;
  font-weight: 700;
  margin-top: 5px;
  min-height: 18px;
  padding-left: 4px;
  display: flex;
  align-items: center;
  gap: 5px;
}
.msg.error { color: #ef4444; }
.msg.ok    { color: #10b981; }

button[type="submit"] {
  width: 100%;
  padding: 13px;
  border: none;
  border-radius: 14px;
  font-size: 16px;
  font-family: 'Nunito', Arial, sans-serif;
  font-weight: 800;
  color: #ffffff;
  background: linear-gradient(135deg, #7c3aed 0%, #db2777 60%, #f59e0b 100%);
```

```
  cursor: pointer;
  margin-top: 0.5rem;
  letter-spacing: 0.3px;
  transition: opacity 0.2s, transform 0.1s;
}
button[type="submit"]:hover  { opacity: 0.90; }
button[type="submit"]:active { transform: scale(0.98); }
.success-msg {
  display: none;
  margin-top: 1rem;
  padding: 14px 16px;
  border-radius: 14px;
  background: linear-gradient(135deg, #d1fae5, #a7f3d0);
  border: 2px solid #6ee7b7;
  font-size: 14px;
  font-weight: 700;
  color: #065f46;
  text-align: center;
}
```

Step 3:

Adding JavaScript for Validation Finally, add JavaScript to validate the input fields using Validator.js and display error messages.

JAVASCRIPT

```
var ERR_ICON = '<svg width="13" height="13" viewBox="0 0 16 16" fill="none">'
  + '<circle cx="8" cy="8" r="7" fill="#fee2e2"/>'
  + '<path d="M5.5 5.5l5 5M10.5 5.5l-5 5" stroke="#ef4444" stroke-width="2" stroke-linecap="round"/>'
  + '</svg>';
var OK_ICON = '<svg width="13" height="13" viewBox="0 0 16 16" fill="none">'
  + '<circle cx="8" cy="8" r="7" fill="#d1fae5"/>'
  + '<path d="M4.5 8l2.5 2.5L11.5 5" stroke="#10b981" stroke-width="2" stroke-linecap="round" stroke-linejoin="round"/>'
  + '</svg>';
var rules = {
  name: function (v) {
    if (!v.trim()) return { ok: false, msg: 'Name is required.' };
    if (v.trim().length < 2) return { ok: false, msg: 'At least 2 characters please.' };
```

```
    if (!/^[a-zA-Z\s'\-]+$/.test(v.trim())) return { ok: false, msg: 'Letters, spaces, hyphens & apostrophes only.' };
    return { ok: true, msg: 'Perfect name!' };
  },
  email: function (v) {
    if (!v.trim()) return { ok: false, msg: 'Email address is required.' };
    if (!validator.isEmail(v)) return { ok: false, msg: "That doesn't look like a valid email." };
    return { ok: true, msg: 'Valid email address!' };
  },
  phone: function (v) {
    if (!v.trim()) return { ok: false, msg: 'Phone number is required.' };
    if (!validator.isMobilePhone(v.replace(/\s/g, ''), 'any', { strictMode: false }))
      return { ok: false, msg: 'Please enter a valid phone number.' };
    return { ok: true, msg: 'Valid phone number!' };
  },
  url: function (v) {
    if (!v.trim()) return { ok: true, msg: '' };
    if (!validator.isURL(v, { require_protocol: true }))
      return { ok: false, msg: 'URL must start with http:// or https://' };
    return { ok: true, msg: 'Valid website URL!' };
  }
};
var validState = { name: false, email: false, phone: false, url: true };
function showMsg(msgId, text, isError) {
  var el = document.getElementById(msgId);
  if (!text) {
    el.innerHTML = '';
    el.className = 'msg';
    return;
  }
  el.className = isError ? 'msg error' : 'msg ok';
  el.innerHTML = (isError ? ERR_ICON : OK_ICON) + text;
}
function setInputState(inputId, isValid) {
  var el = document.getElementById(inputId);
  el.classList.toggle('valid', isValid);
  el.classList.toggle('invalid', !isValid);
```

```
}
function updateProgress() {
  var count = Object.values(validState).filter(Boolean).length;
  var pct = Math.round((count / 4) * 100);
  document.getElementById('pfill').style.width = pct + '%';
  document.getElementById('pct').textContent = pct + '%';
}
var fields = ['name', 'email', 'phone', 'url'];
fields.forEach(function (id) {
  document.getElementById(id).addEventListener('input', function () {
    var value = this.value;
    if (id === 'url' && !value.trim()) {
      this.classList.remove('valid', 'invalid');
      showMsg('urlMsg', '', false);
      validState.url = true;
      updateProgress();
      return;
    }
    var result = rules[id](value);
    setInputState(id, result.ok);
    showMsg(id + 'Msg', result.msg, !result.ok);
    validState[id] = result.ok;
    updateProgress();
  });
});
document.getElementById('myForm').addEventListener('submit', function (e) {
  e.preventDefault();
  var allValid = true;
  fields.forEach(function (id) {
    var value = document.getElementById(id).value;
    if (id === 'url' && !value.trim()) {
      document.getElementById(id).classList.remove('valid', 'invalid');
      showMsg('urlMsg', '', false);
      validState.url = true;
      return;
    }
```

```
    var result = rules[id](value);
    setInputState(id, result.ok);
    showMsg(id + 'Msg', result.msg, !result.ok);
    validState[id] = result.ok;
    if (!result.ok) allValid = false;
  });

  updateProgress();
  var successMsg = document.getElementById('successMsg');
  if (allValid) {
    successMsg.style.display = 'block';
    console.log('Name:', document.getElementById('name').value);
    console.log('Email:', document.getElementById('email').value);
    console.log('Phone:', document.getElementById('phone').value);
    console.log('URL:', document.getElementById('url').value);
    setTimeout(function () { successMsg.style.display = 'none'; }, 4000);
  } else {
    successMsg.style.display = 'none';
  }
});
```

OUTPUT

✦ Get in Touch

Fill in the form below — we'd love to hear from you!

75%

1 FULL NAME

nethra

Perfect name!

2 EMAIL ADDRESS

nethra@gmail.com

Valid email address!

3 PHONE NUMBER

7876

Include country code (e.g. +91 for India, +1 for US)

Please enter a valid phone number.

4 WEBSITE optional

https://yoursite.com

Send Message ✦